

Algebra

Tuesday, 3 June, 2025

8:26 AM

I: ALGEBRAIC MANIPULATIONS

$$xy = 2, \quad x + y = 4$$
$$x^2 + y^2$$

* don't solve for variables

$$(x+y)^2 = 4^2$$
$$x^2 + \underbrace{2xy} + y^2 = 16$$
$$x^2 + 2(2) + y^2 = 16$$
$$x^2 + y^2 = 12$$

$$143x - 77y = 451$$
$$\left(\begin{array}{l} 299x - 161y = 23(13x - 7y) \\ = 23(41) \end{array} \right.$$
$$\rightarrow 13x - 7y = 41$$

$$x - \frac{1}{x} = 4 \rightarrow x^2 - 2(x)(\frac{1}{x}) + \frac{1}{x^2} = 16$$
$$x^2 + \frac{1}{x^2} \quad \quad \quad x^2 - 2 + \frac{1}{x^2} = 16$$
$$\quad \quad \quad \quad \quad \quad \quad x^2 + \frac{1}{x^2} = 18$$

$$x + \frac{1}{y} = 4$$

$$y + \frac{1}{z} = 1$$

$$xyz = ?$$

$$z + \frac{1}{x} = \frac{7}{3}$$

$$(x + \frac{1}{y})(y + \frac{1}{z})(z + \frac{1}{x}) = \frac{28}{3}$$

$$xyz + y + x + \frac{1}{z} + z + \frac{1}{x} + \frac{1}{y} + \frac{1}{xyz} = \frac{28}{3}$$

$$xyz + x + y + z + \frac{1}{x} + \frac{1}{y} + \frac{1}{z} + \frac{1}{xyz} = \frac{28}{3}$$

$$xyz + \underbrace{(x + \frac{1}{y})}_4 + \underbrace{(y + \frac{1}{z})}_1 + \underbrace{(z + \frac{1}{x})}_{\frac{7}{3}} + \frac{1}{xyz} = \frac{28}{3}$$

$$xyz + \frac{1}{xyz} = 2$$

$$a = xyz = 1 \rightarrow \frac{a^2}{a} + \frac{1}{a} = 2$$

$$\frac{a^2+1}{a} = 2$$

$$a^2+1 = 2a$$

$$a^2 - 2a + 1 = 0$$

$$(a-1)^2 = 0$$

$$a-1 = 0$$

$$a = 1$$

$$(b-196a)^2 - 128(b-14c)(c-14a) = 0$$

$$\max \frac{b-14c}{c-14a} = x$$

$$c-14a = y$$

$$b-196a = x + 14y$$

$$x + 14y = (b-14c) + 14(c-14a)$$

$$= b - \cancel{14c} + \cancel{14c} - 14^2 a$$

$$= b - 196a$$

$$\max \frac{x}{y}$$

$$(x+14y)^2 - 128xy = 0$$

$$x^2 + \underline{28xy} + 196y^2 - \underline{128xy} = 0$$

$$x^2 - 100xy + 196y^2 = 0$$

$$\frac{x^2}{y^2} - \frac{100xy}{y^2} + \frac{196y^2}{y^2} = \frac{0}{y^2}$$

$$\left(\frac{x}{y}\right)^2 - 100\left(\frac{x}{y}\right) + 196 = 0 \quad z = \frac{x}{y}$$

$$\left[\begin{array}{l} z^2 - 100z + 196 = 0 \\ \rightarrow (z-98)(z-2) = 0 \\ z = 2 \text{ or } \underline{z = 98} \end{array} \right.$$

quadratic crash course !!

$$ax^2 + bx + c = 0$$

0/1/2 real solutions

① quad formula

② factorisation

$$(\underline{x-p})(\underline{x-q}) = 0$$

$$x=p \text{ or } x=q$$

$$x^2 - 4x + 3 = 0$$

$$(\underline{x-3})(\underline{x-1}) = 0$$

$$\begin{array}{lcl} x-3=0 & \text{or} & x-1=0 \\ x=3 & & x=1 \end{array}$$

2: TRIVIAL INEQUALITY

$$x^2 \geq 0, x \in \mathbb{R}$$

$$x^2 = 0 \Rightarrow x = 0$$

$$\begin{array}{c} (\underline{x-2025})^2 = 0 \\ \parallel \\ 0 \\ x = 2025 \end{array}$$

$$\begin{array}{c} 22(\underline{x-22})^{\underline{22}} + 24(\underline{y-24})^{\underline{24}} \\ \parallel \qquad \qquad \parallel \\ 0 \qquad \qquad \qquad 0 \\ + 26(\underline{z-26})^{\underline{26}} = 0 \\ \parallel \\ 0 \end{array}$$

minimum

$$x^2 + 20x + 25$$

$$\begin{array}{l} \min = -75 \\ \text{at } x = -10 \end{array}$$

$$\begin{aligned}
 & \text{at } x = -10 \\
 & \boxed{\begin{aligned} & ! \quad (x+10)^2 = x^2 + 20x + 100 \\ & \rightarrow \quad \underbrace{(x+10)^2}_{\geq 0} - 75 \geq -75 \end{aligned}} \\
 & \quad \quad \quad x = -10
 \end{aligned}$$

$$\begin{aligned}
 & \min \quad (xy-1)^2 + (x+y)^2 \\
 & \quad \quad \quad xy-1=0, \quad x+y=0 \quad \times \\
 & \quad \quad \quad x^2y^2 - 2xy + 1 + x^2 + 2xy + y^2 \\
 & \quad \quad \quad = \underbrace{x^2y^2}_{\geq 0} + 1 + \underbrace{x^2}_{\geq 0} + \underbrace{y^2}_{\geq 0} \geq 1 \\
 & \quad \quad \quad x=y=0
 \end{aligned}$$

Prove $a^2 + b^2 \geq 2ab$

$$a, b \in \mathbb{R}$$

$$(a-b)^2 \geq 0$$

$$\Rightarrow a^2 - 2ab + b^2 \geq 0$$

$$\Rightarrow a^2 + b^2 \geq 2ab.$$

3: AM-GM INEQ.

AM-GM INEQUALITY

$\nwarrow$ arithmetic mean
 $\swarrow$ "mean"
 $\rightarrow$ tmh smala, bhy blgn

$\nwarrow$ geometric mean
 $\rightarrow$ drb smala, power $1/blgn$

$$a_1, a_2, \dots, a_n \text{ are positive real numbers}$$

$$a, b \rightarrow GM = (ab)^{1/2} = \sqrt{ab} \quad x^{1/2} = \sqrt{x}$$

$$a, b, c \rightarrow GM = \sqrt[3]{abc}$$

$$\mathbb{R}^+, AM \geq GM$$

$$a, b, c \in \mathbb{R}^+$$

$$a, b \in \mathbb{R}^+$$

?

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

* equality iff variables equal

$$\frac{a+b}{2} = \sqrt{ab} \Leftrightarrow a=b$$

$$2 \rightarrow \frac{x+y}{2} \geq \sqrt{xy}$$

Prove by $a^2+b^2 \geq 2ab$

AM - GM

$$\frac{a^2+b^2}{2} \geq \sqrt{a^2b^2} = ab$$

$$\Rightarrow a^2+b^2 \geq 2ab.$$

$x + \frac{1}{x} \geq 2$, $x \in \mathbb{R}^+$
and find equality case

$$\frac{x+1}{2} \geq \sqrt{x(1)}$$

$$\hookrightarrow x+1 \geq 2\sqrt{x}$$

$$\frac{a+b}{2} \geq \sqrt{ab}$$

$$\frac{x + \frac{1}{x}}{2} \geq \sqrt{x\left(\frac{1}{x}\right)} = 1$$

$$x^2 + 2x + 1 \geq 4x$$

$$x^2 + 1 \geq 2x$$

$$x + 1/x \geq 2$$

$$x + \frac{1}{x} \geq 2$$

Prove $(a+b)^2 \geq 4ab$

$$\frac{a+b}{2} \geq \sqrt{ab} \Rightarrow a+b \geq 2\sqrt{ab}$$

$$(a+b)^2 \geq (2\sqrt{ab})^2 = 4ab$$

Prove $a^2 + b^2 + c^2 \geq ab + bc + ca$

$$\frac{a^2+b^2}{2} \geq ab, \quad \frac{b^2+c^2}{2} \geq bc,$$

$$\frac{a^2+c^2}{2} \geq ac$$

$$+mbh \text{ sum } a \rightarrow \underbrace{\frac{a^2+b^2}{2} + \frac{b^2+c^2}{2} + \frac{a^2+c^2}{2}}_{= a^2+b^2+c^2} \geq ab+bc+ac$$

Prove $a^3 + b^3 \geq ab(a+b)$

$$= a^2b + ab^2$$

$$a^3 + b^3 = (a+b)(\underbrace{a^2+b^2}_{\geq 2ab} - ab) \geq (a+b)(2ab - ab) = ab(a+b)$$

$$+ \frac{a^3 + a^3 + b^3}{3} \geq \sqrt[3]{a^3 a^3 b^3} = a^2b$$

$$+ \frac{a^3 + b^3 + b^3}{3} \geq ab^2$$

equality
 $a=b$

$$a^3 + b^3 \geq a^2b + ab^2$$

$$(\underline{a+b})(\underline{b+c})(\underline{c+a}) \geq 8abc$$

$$\frac{a+b}{2} \geq \sqrt{ab}, \quad \frac{b+c}{2} \geq \sqrt{bc}, \quad \frac{c+a}{2} \geq \sqrt{ac}$$

$$\left(\frac{a+b}{2}\right)\left(\frac{b+c}{2}\right)\left(\frac{c+a}{2}\right) \geq \sqrt{a^2b^2c^2} = abc$$

$$\frac{(a+b)(b+c)(c+a)}{8} \geq abc$$

$$(a+b)(b+c)(c+a) \geq 8abc$$

$$\left(\underbrace{a_1 + a_2 + \dots + a_n}_{\downarrow}\right) \left(\underbrace{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}\right) \geq n^2$$

$$\begin{aligned} \frac{a_1 + a_2 + \dots + a_n}{n} &\geq \sqrt[n]{a_1 a_2 \dots a_n} & \frac{\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}}{n} &\geq \sqrt[n]{\frac{1}{a_1 a_2 \dots a_n}} \\ a_1 + a_2 + \dots + a_n &\geq n \sqrt[n]{a_1 a_2 \dots a_n} & \frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n} &\geq n \sqrt[n]{\frac{1}{a_1 a_2 \dots a_n}} \end{aligned}$$

$$\begin{aligned} &(a_1 + a_2 + \dots + a_n) \left(\frac{1}{a_1} + \frac{1}{a_2} + \dots + \frac{1}{a_n}\right) \\ &\geq \left(n \sqrt[n]{a_1 a_2 \dots a_n}\right) \left(n \sqrt[n]{\frac{1}{a_1 a_2 \dots a_n}}\right) = n^2 \end{aligned}$$

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} \geq a + b + c + d$$

$$\frac{b + \frac{a^2}{b}}{2} \geq \sqrt{b \left(\frac{a^2}{b}\right)} = a \quad \frac{d + \frac{c^2}{d}}{2} \geq c$$

$$\frac{c + \frac{b^2}{c}}{2} \geq b \quad \frac{a + \frac{d^2}{a}}{2} \geq d$$

$$\frac{\cancel{a+b+c+d} + \frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a}}{2} \geq a+b+c+d$$

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} \geq a+b+c+d$$

$$\frac{a^2}{b} + \frac{b^2}{c} + \frac{c^2}{d} + \frac{d^2}{a} \geq a + b + c + d$$

$$\underline{abc} = 1 \rightarrow \underline{a^{1/3} b^{1/3} c^{1/3}} = 1$$

$$\rightarrow \text{prove } a^2 + b^2 + c^2 \geq a + b + c$$

$$\begin{aligned} \frac{a^2 + a^2 + a^2 + a^2 + b^2 + c^2}{6} &\geq \sqrt[6]{a^8 b^2 c^2} = \underbrace{a^{4/3} b^{1/3} c^{1/3}}_{= a^{1/3} b^{1/3} c^{1/3} \cdot 4:1:1} \\ &+ \underbrace{a^{1/3} b^{4/3} c^{1/3}}_{= a^{1/3} b^{1/3} c^{4/3}} \end{aligned}$$

$$+ \frac{b^2 + b^2 + b^2 + b^2 + a^2 + c^2}{6} \geq a^{1/3} b^{4/3} c^{1/3}$$

$$+ \frac{c^2 + c^2 + c^2 + c^2 + a^2 + b^2}{6} \geq a^{1/3} b^{1/3} c^{4/3}$$

Austrian JMO 2023 P1

$$x, y, z \neq 0$$

$$\frac{x+y}{z} = \frac{y+z}{x} = \frac{z+x}{y} = k$$

$$k^3 = \frac{(x+y)(y+z)(z+x)}{xyz} = ?$$

$$\rightarrow x+y = kz$$

$$y+z = kx$$

$$z+x = ky$$

$$\rightarrow 2(x+y+z) = k(x+y+z)$$

$$\downarrow (2-k)(x+y+z) = 0$$

$$\swarrow x+y+z \neq 0$$

$$\underline{k=2}$$

$$k^3 = 8$$

$$\checkmark$$

$$\searrow x+y+z = 0$$

$$\frac{\overset{-z}{(x+y)} \overset{-x}{(y+z)} \overset{-y}{(z+x)}}{xyz}$$

$$= \frac{(-z)(-x)(-y)}{xyz} = -1$$

$$k^3 = 8 \\ \checkmark \\ x=y=z$$

$$\frac{xyz}{(-z)(-x)(-y)} = \frac{-1}{-1} = 1$$

$$x=y=1, z=-2$$

Turkey JMO 2020 P1

$$2x^2 + y^2 + 7 = 2(x+1)(y+1)$$

find all possible
values of (x, y)

$$2x^2 + y^2 + 7 = 2(xy + x + y + 1) \\ = 2xy + 2x + 2y + 2$$

$$\frac{2x^2 + y^2}{x^2 + x^2} - \frac{2xy - 2x - 2y + 5}{1^2 + 2^2} = 0$$

$$\frac{(x-2)^2 + x^2 + y^2 - 2xy + 2x - 2y + 1}{x^2 - 4x + 4} = 0 \\ = \frac{(x-y+1)^2}{(x-y+1)^2} \\ = (x-y+1)(x-y+1) \\ = x^2 - xy + x - xy + y^2 - y + x - y + 1 \\ = x^2 + y^2 - 2xy + 2x - 2y + 1$$

$$\rightarrow \frac{(x-2)^2}{0} + \frac{(x-y+1)^2}{0} = 0$$

$$x=2, y=3$$

JBMO SL 2014 A2

$$a, b, c \in \mathbb{R}^+$$

$$abc = 1/8$$

$$a, b, c \in \mathbb{R}^+ \text{ and } a+b+c=15,$$

$$\underbrace{a^2 + b^2 + c^2} + \underbrace{a^2b^2 + b^2c^2 + c^2a^2} \geq \frac{15}{16}$$

$$\frac{a^2 + b^2 + c^2}{3} \geq \sqrt[3]{a^2b^2c^2} = \sqrt[3]{1/8^2} = 1/4$$

$$\rightarrow a^2 + b^2 + c^2 \geq 3/4$$

$$\frac{a^2b^2 + b^2c^2 + c^2a^2}{3} \geq \sqrt[3]{a^4b^4c^4} = \sqrt[3]{1/8^4} = \frac{1}{16}$$

$$\rightarrow a^2b^2 + b^2c^2 + c^2a^2 \geq 3/16$$

$$\sim \geq 3/4 + 3/16 = 12/16 + 3/16 = 15/16.$$

Al-khwarizmi IMO 2024 P6

$$a = b \neq c \neq a$$

$$a + b + c = 0$$

$$a^2 - b = b^2 - c = c^2 - a = k$$

find all possible values of $ab + ac + bc$

hint: difference of squares

$$a^2 - b = b^2 - c$$

$$\underbrace{a^2 - b^2}_{(a-b)(a+b)} = b - c$$

$$\times \begin{cases} (a-b)(a+b) = b-c \\ (b-c)(b+c) = c-a \\ (c-a)(c+a) = a-b \end{cases}$$

$$\underbrace{(a+b)}_{-c} \underbrace{(b+c)}_{-a} \underbrace{(c+a)}_{-b} = 1$$

$$abc = -1$$

$$a + b + c = 0$$

→ well-known

$$\Rightarrow a^3 + b^3 + c^3 = 3abc = 3(-1) = -3$$

$$a^3 + b^3 + c^3 = (a+b+c)(a^2+b^2+c^2-ab-bc-ca) + 3abc$$

$$\begin{aligned} a^3 + b^3 + c^3 - (ab+bc+ca) &= 0 \\ &= (a^3 - ab) + (b^3 - bc) + (c^3 - ca) \\ &= a(\underbrace{a^2 - b}) + b(\underbrace{b^2 - c}) + c(\underbrace{c^2 - a}) \\ &= ak + bk + ck \\ &= k(\underbrace{a+b+c}) \\ &= 0 \end{aligned}$$

$$\therefore ab+bc+ca = a^3 + b^3 + c^3 = -3$$